Hussain Raza

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Education

University of Virginia Charlottesville, VA

Bachelor of Science in Computer Science, AI Concentration | Minor in Data Science *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms I & II, Computer Systems & Organization I & II, Mobile Application Development, Software Engineering, Cybersecurity, Discrete Mathematics, Data Science

Experience

Amazon Web Services (AWS) May 2025 – August 2025

Software Development Engineer Intern, CloudWatch Agent Team *Arlington, VA*

- Made failures visible for 3 agent and GPU monitoring components by emitting health events and status conditions through the Kubernetes API
- Enabled Kubernetes to automatically detect and restart failing pods across 4 CloudWatch Agent Operator components by implementing liveness and readiness probes in Go
- Merged 3 pull requests (about 700 lines of Go, including unit tests) into 2 public AWS open-source repositories after code review by the CloudWatch Agent team
- Used Kiro, AWS's AI coding IDE, to map an unfamiliar Go codebase and draft changes, then reviewed and tested all AI-generated code before code review

MIT Introduction to Technology, Engineering, and Science (MITES) June 2023 – January 2024

Student *Virtual*

- Selected as 1 of ∼230 students from ∼2,700 applicants for MIT's 6-month STEM program
- Used ArcGIS Online to analyze relationships between socioeconomic factors and healthcare access
- Presented geographic visualizations and findings at a research symposium

Projects

Skypane | *Swift, SwiftUI, SpriteKit, Apple Foundation Models* Fall 2026

- Developed and released a native macOS weather application using Swift, SwiftUI, and SpriteKit, with live weather visualization and window-aware particle animations
- Integrated on-device generative AI using Apple's Foundation Models to create custom scenes from natural-language prompts, with validation that limits generated effects to user requests
- Optimized rendering to reduce thunderstorm-scene CPU utilization from 35% to 18% and idle memory usage by 45%, suspending rendering when inactive

UVA CIO Management Platform | *Python, Django, JavaScript, PostgreSQL, AWS S3* Spring 2026

- Delivered direct messaging with file attachments and per-team group chat in a Django web app for a 5-person Agile team by building the data models, views, and JavaScript UI that sends messages without page reloads
- Gathered and triaged feedback from 11 external beta testers across 3 roles into tracked GitHub issues as QA lead
- Fixed a Google sign-in authentication bug that overwrote profiles, 2 tester-reported bugs, and a failing CI build

Missile Booster Optimization Simulator | *Python, Tkinter, Matplotlib* Spring 2025

- Achieved under 2% simulation error across 24 validated test cases by building a Python simulator for a U.S. Navy client comparing delta-V and range for 3-stage vs. pop-out booster designs
- Let users compare designs without code by adding a Tkinter GUI with matplotlib charts and CSV export

Technical Skills and Honors

Languages: Python, Java, Go, Swift, JavaScript, SQL, C, R
Frameworks: SwiftUI, SpriteKit, Django, R Shiny
Developer Tools: AWS (CloudWatch, Bedrock), Docker, Kubernetes, Git, GitHub Actions, Linux, Xcode
Libraries: pandas, NumPy, Matplotlib
AI Tools: Claude Code, Codex, Kiro, Apple Foundation Models
Honors: APIA Scholar, Amazon Future Engineer Scholar, QuestBridge National College Match Finalist